

A Spatial Segregation Measurement Package in R

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Updating urban segregation computation with R, TUDelft

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Background

Reardon and colleague's segregation measures continue to represent an important paradigm in segregation measurement in sociology.

But these measures remain underutilized because of a lack of tools. The seg package was the main practical tool, but it had a number of missing features:

- Lack of support for continuous and ordinal varialbes.
- Incomplete implementation of pycnophylactic interpolation.
- No built-in support for k-nearest neighbors approaches.
- No built-in support for network data.

Introducing spseg

- A C++ engine based on robust Boost.org libraries.
- Modernized spatial data handling with `sf`.
- Radial (Reardon et al.) and knn (Östh) neighbor search.
- Return of spatially weighted local environments.
- Calculation of multigroup and pairwise indices.
- Exposure and isolation measures.
- Support for micro-and aggregated data.
- Generates measures for categorical, ordinal, and continuous covariates.
- New pycnophylactic interpolation procedures.
- Support for network data (under development)

Local population and regional weights

$$p_{ig} = \frac{\sum_j \lambda_{ij} N_{jg}}{\sum_h \sum_j \lambda_{ij} N_{jh}}, \qquad P_g = \frac{\sum_i N_{ig}}{T}, \qquad \alpha_i = \frac{T_i}{T}$$

$$J_f = 1 - \frac{\sum_i \alpha_i v_{i,f}}{v_f}, \qquad D = \frac{\sum_i \alpha_i d_i}{d_0}$$

[1] Reardon & O'Sullivan (2004), *Measures of Spatial Segregation*, Sociological Methodology 34:121–162.

Binary variation functions

$$\phi_H(p) = -p \log p - (1-p) \log(1-p),$$

$$\phi_R(p) = p(1-p),$$

$$\phi_S(p) = 2\sqrt{p(1-p)}.$$

Cutoffs and ranks

$$C_{ik} = \sum_{g \le k} p_{ig}, \qquad Q_k = \sum_{g \le k} P_g, \qquad \langle a_i \rangle = \sum_i \alpha_i a_i$$

$$\mathcal{J}_O[\phi] = 1 - \frac{\langle \sum_k \phi(C_{ik}) \rangle}{\sum_k \phi(Q_k)} \qquad \mathcal{J}_C[\phi] = 1 - \frac{\langle \int_0^1 \phi(F_i(u))\, du \rangle}{\int_0^1 \phi(F(u))\, du}$$

$$\mathcal{D}_O = \frac{\langle \sum_k |C_{ik} - Q_k| \rangle}{2 \sum_k Q_k(1 - Q_k)}, \qquad \mathcal{D}_C = \frac{\langle \int_0^1 |F_i(u) - F(u)|\, du \rangle}{2 \int_0^1 F(u)(1 - F(u))\, du}$$

Measures by data type

Measure	Categorical[1]	Ordinal[2]	Continuous[3]
Dissimilarity D	$\frac{\langle \sum_g \lvert p_{ig} - P_g \rvert \rangle}{2 \sum_g P_g(1-P_g)}$	$\mathcal{D}_O^\dagger$	$\mathcal{D}_C^\dagger$
Information H	$1 - \frac{\langle E(\mathbf{p}_i) \rangle}{E(\mathbf{P})}$	$\mathcal{J}_O[\phi_H]$	$\mathcal{J}_C[\phi_H]$
Diversity R	$1 - \frac{\langle I(\mathbf{p}_i) \rangle}{I(\mathbf{P})}$	$\mathcal{J}_O[\phi_R]$	$\mathcal{J}_C[\phi_R]$
Root S[4]	$1 - \frac{\langle G(\mathbf{p}_i) \rangle}{G(\mathbf{P})}$	$\mathcal{J}_O[\phi_S]$	$\mathcal{J}_C[\phi_S]$

$E(\mathbf{p}) = -\sum_g p_g \log p_g$; $I(\mathbf{p}) = \sum_g p_g(1-p_g)$; $G(\mathbf{p}) = (\prod_{g:P_g>0} p_g)^{1/K}$, $K = \#\{g : P_g > 0\}$.

[1] Reardon & Firebaugh (2002), *Measures of Multigroup Segregation*, Sociological Methodology 32:33–67.

[2] Reardon (2009), *Measures of Ordinal Segregation*, Research on Economic Inequality 17:129–155.

[3] Reardon (2011), *Measures of Income Segregation*, Stanford working paper, revised September. Empirical/interpolation conventions follow spseg.

[4] Hutchens (2004), Int. Econ. Review 45:555–578 (binary root); Frankel & Volij (2011), J. Econ. Theory 146:1–38 (multigroup Atkinson).

Continuous input: exact values and intervals

Exact values

$$z_g = \sum_{h<g} P_h + P_g/2$$

Empirical CDFs · Weighted midranks

Interval counts

$$G_i(q) = C_{i,k-1} + \frac{q - Q_{k-1}}{Q_k - Q_{k-1}}(C_{ik} - C_{i,k-1})$$

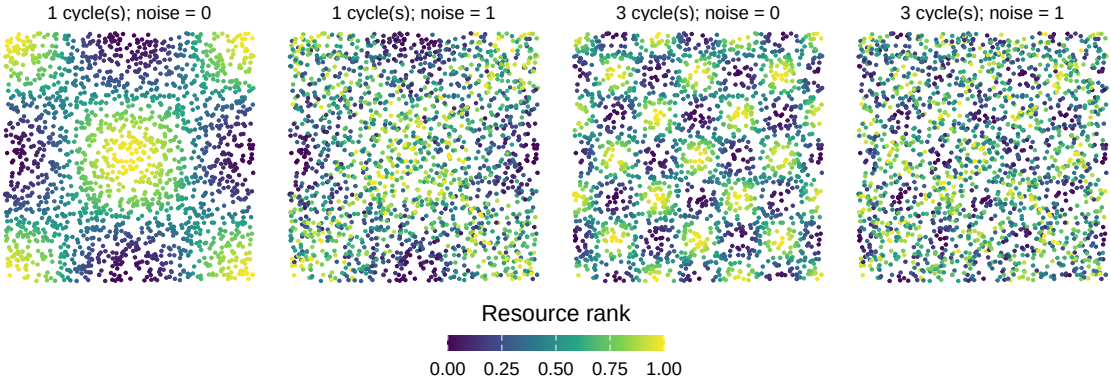
Linear rank CDF · $G(q) = q$

Installation and inputs

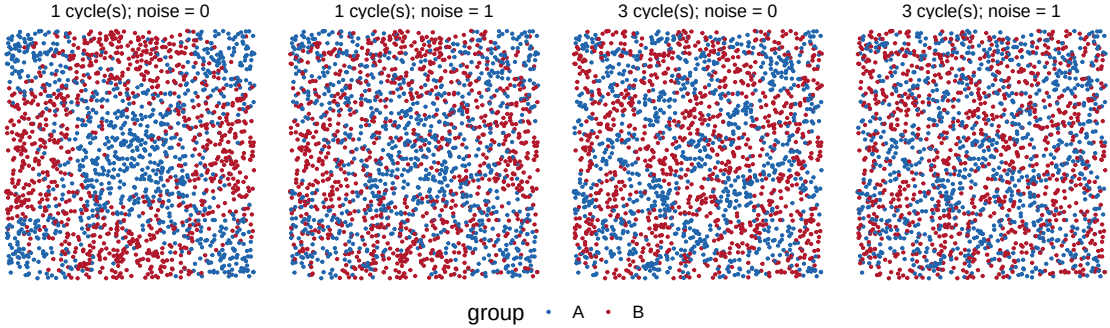
```
install.packages("pak")
pak::pak("bfjarvis/spseg")

library(spseg)
library(sf)
library(tidyverse)
```

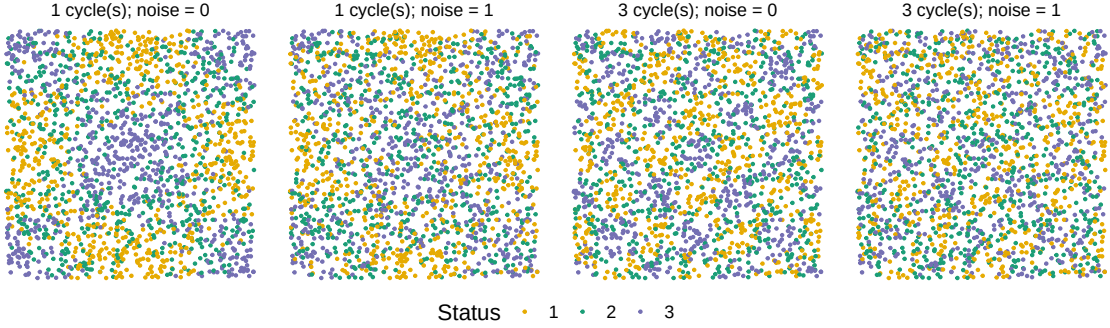
Four residential patterns based on



Correlated group membership



And correlated status



spseg syntax

```
spseg(
  x,
  variables,
  coords = NULL,
  measures = "all",
  bands = NULL,
  weighting = c("biweight", "unweighted", "inverse", "exponential"),
  power = 3,
  normalize = TRUE,
  neighbors = c("radius", "knn"),
  search = c("boost", "brute"),
  surface = c("raw", "grid", "pycno"),
  scope = c("both", "overall", "pairwise"),
  output = c("indices", "full", "localenv"),
  keep_scores = FALSE,
  ...
)
```

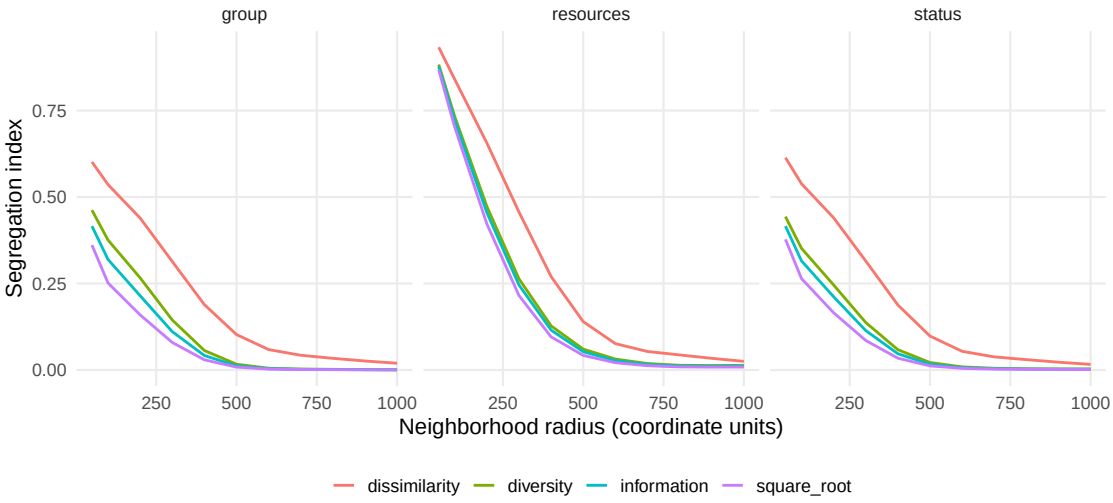
Specifying the `variables` option

```
micro_variables <- list(
  group = spseg_categorical(vars = "group"),
  status = spseg_ordinal(vars = "status"),
  resources = spseg_continuous(
    vars = "resources",
    summary = list(type = "shares", probs = c(.25, .5, .75))
  )
)
```

Calling spseg

```
result <- spseg(
  patterns[[1]],
  coords = c("x", "y"),
  variables = micro_variables,
  bands = c(50, seq(100, 1000, 100)),
  neighbors = "radius",
  weighting = "biweight",
  measures = "all",
  scope = "both",
  output = "full"
)
```

Spatial segregation curves

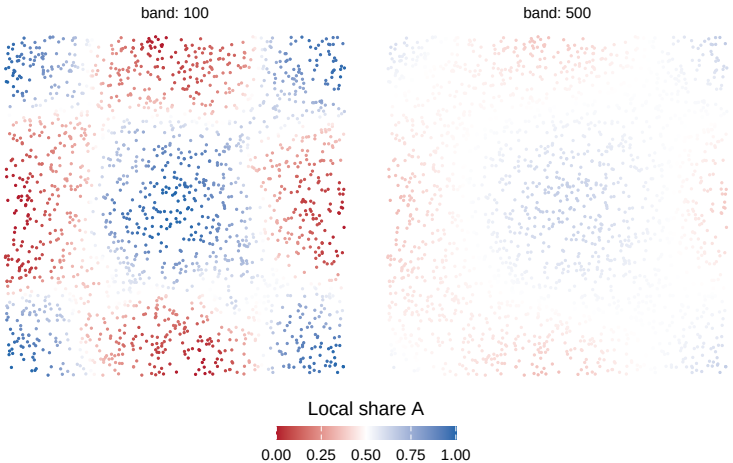


Extract indices and local environments

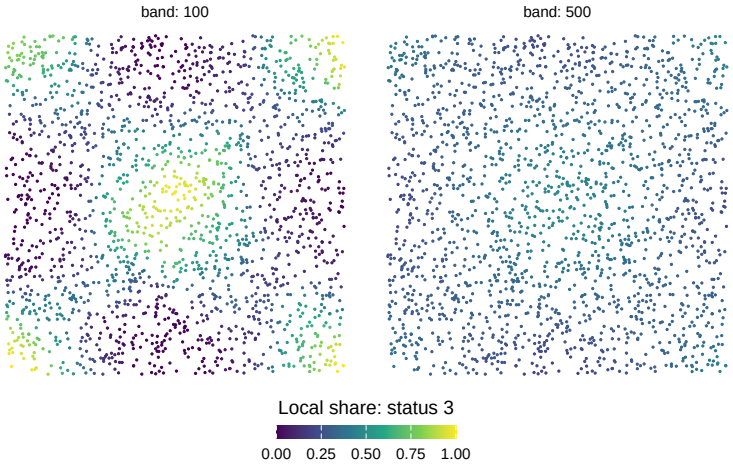
```
indices <- as.data.frame(result, what = "indices")
pairs <- as.data.frame(result, what = "pairwise")

local <- st_as_sf(result, bands = c(100, 500))
local |> filter(variable == "group", category == "A")
```

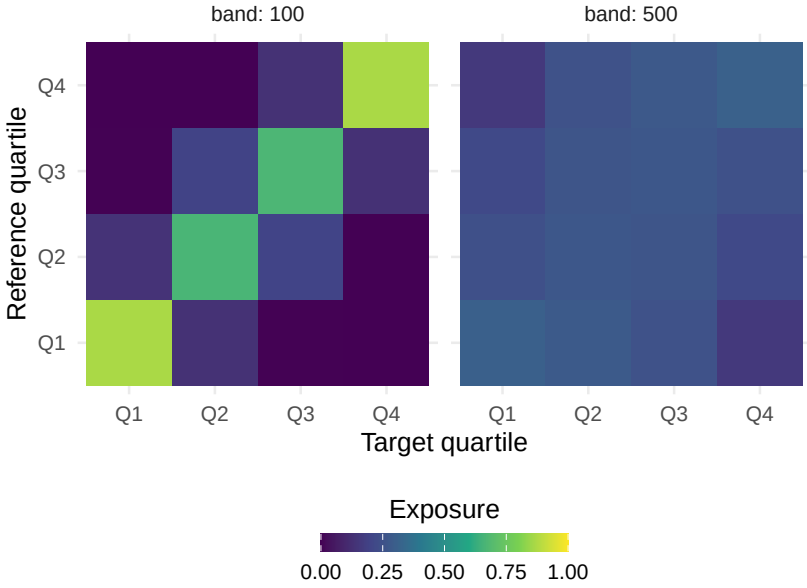
Local group composition



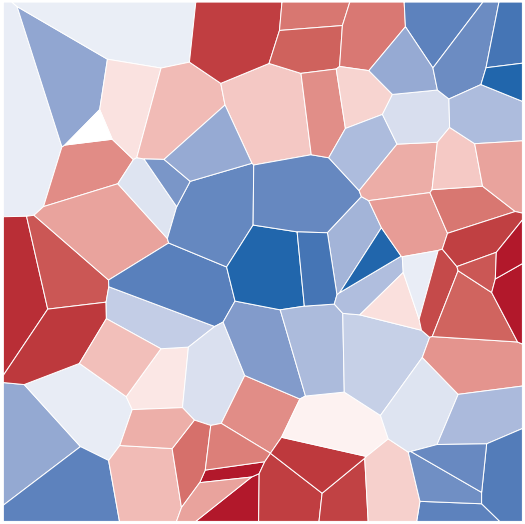
Local environments for ordinal status



Exposure between resource quartiles



Aggregate people into 80 Voronoi areas



Area share A

Select count columns

```
count_variables <- list(
  group = spseg_categorical(vars = c("group_A", "group_B")),
  status = spseg_ordinal(vars = paste0("status_", 1:3)),
  resources = spseg_continuous(
    vars = paste0("resources_", 1:10),
    representation = "intervals"
  )
)
```

Own-area and spatial neighborhoods

```
nonspatial <- spseg(
  voronoi_counts,
  variables = count_variables,
  bands = 0,
  surface = "raw",
  weighting = "unweighted"
)

result_polygons <- spseg(
  voronoi_counts,
  variables = count_variables,
  bands = seq(0, 1000, 100),
  surface = "raw",
  output = "full"
)
```

Choose a population surface

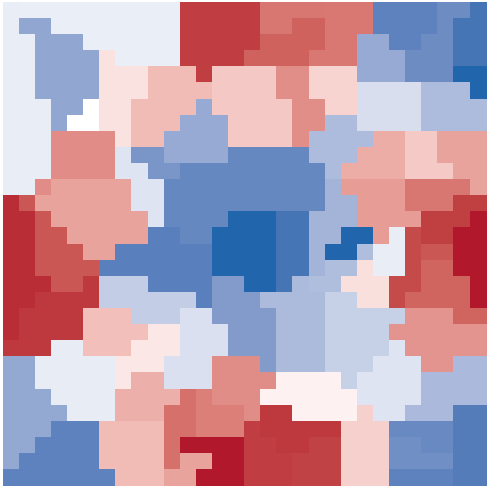
```
result_grid <- spseg(
  voronoi_counts,
  variables = count_variables,
  bands = seq(0, 1000, 100),
  surface = "grid",
  nrow = 30,
  ncol = 30,
  scope = "overall",
  output = "full",
  surface_geometry = "polygons"
)
result_pycno <- spseg(
  voronoi_counts,
  variables = count_variables,
  bands = seq(0, 1000, 100),
  surface = "pycno",
  nrow = 30,
```

Uniform and smoothed population surfaces

Smoothed grid

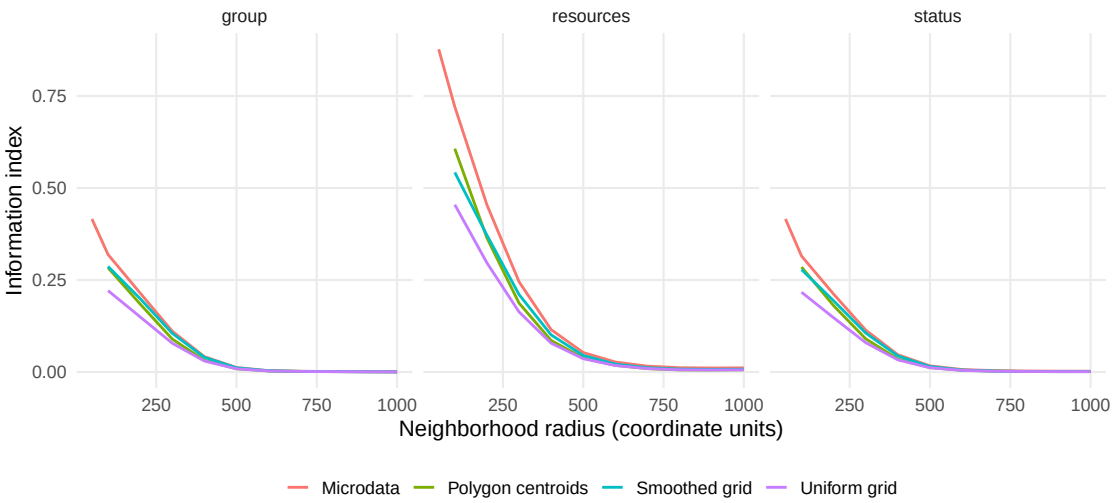


Uniform grid

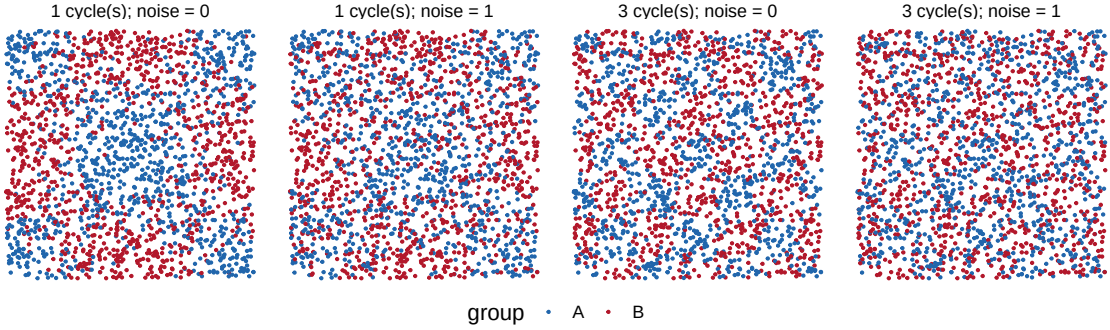


Cell share A

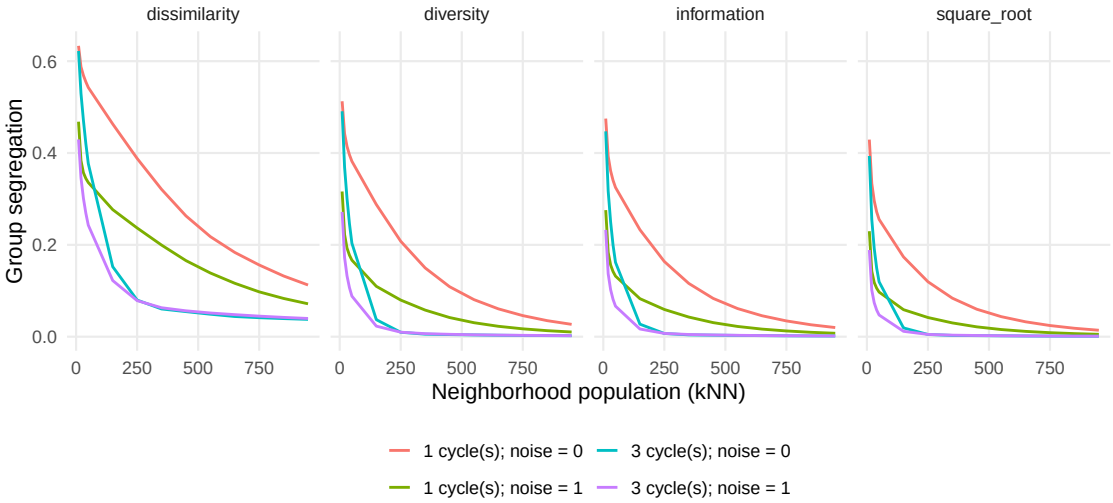
Compare population representations



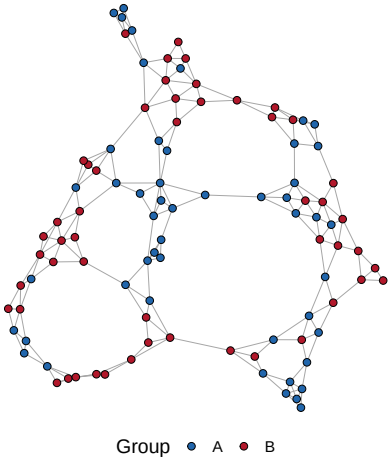
Residential patterns



Compare patterns at population scales



A 100-person population network

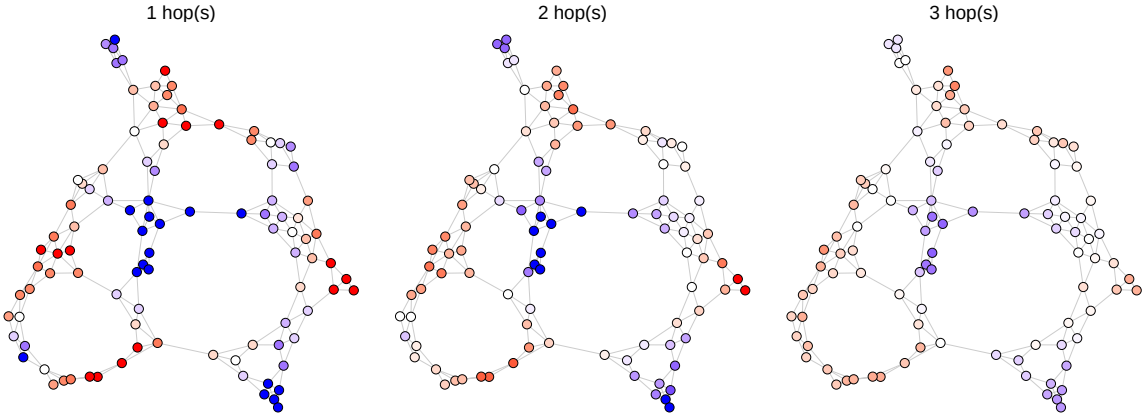


100 nodes · 209 edges

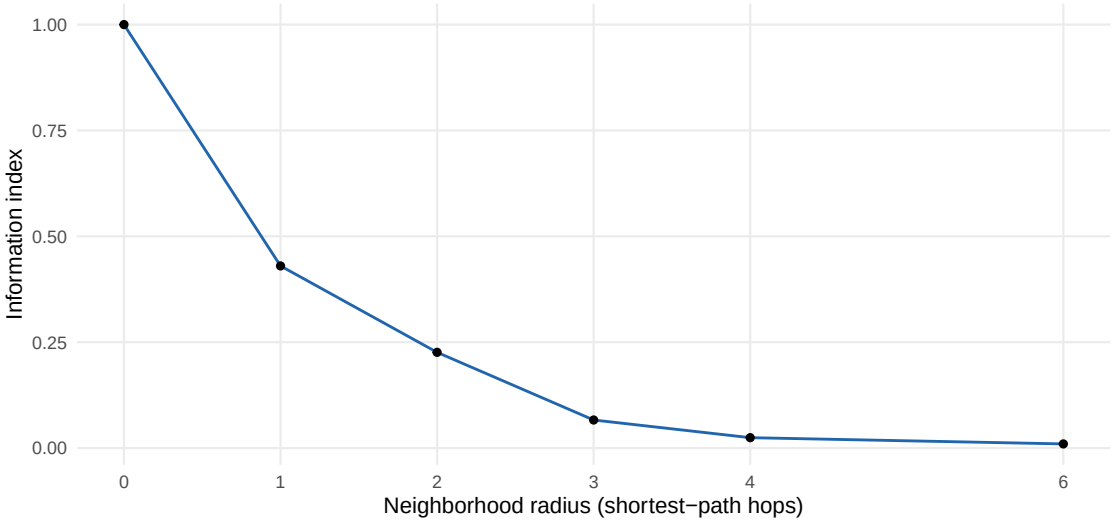
Calculate along network paths

```
network_result <- spseg(
  network_people,
  coords = c("x", "y"),
  variables = list(group = spseg_categorical(vars = "group")),
  network = network,
  node_id = "node",
  bands = c(0, 1, 2, 3, 4, 6),
  weighting = "unweighted",
  output = "full"
)
```

Local environments on the network



Segregation across network scales



Further examples

```
vignette("spseg-getting-started")
vignette("spseg-measure-definitions")
vignette("spseg-networks")
?spseg
```

https://github.com/bfjarvis/spseg